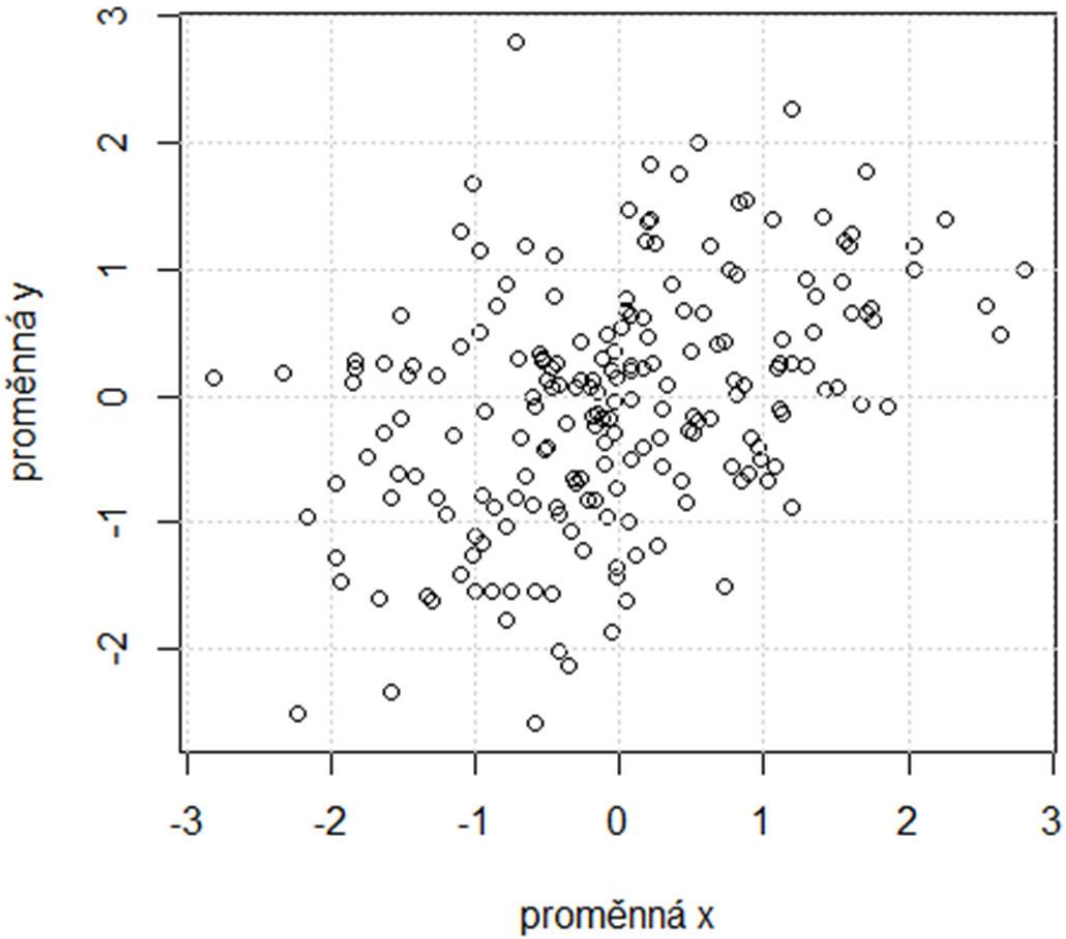
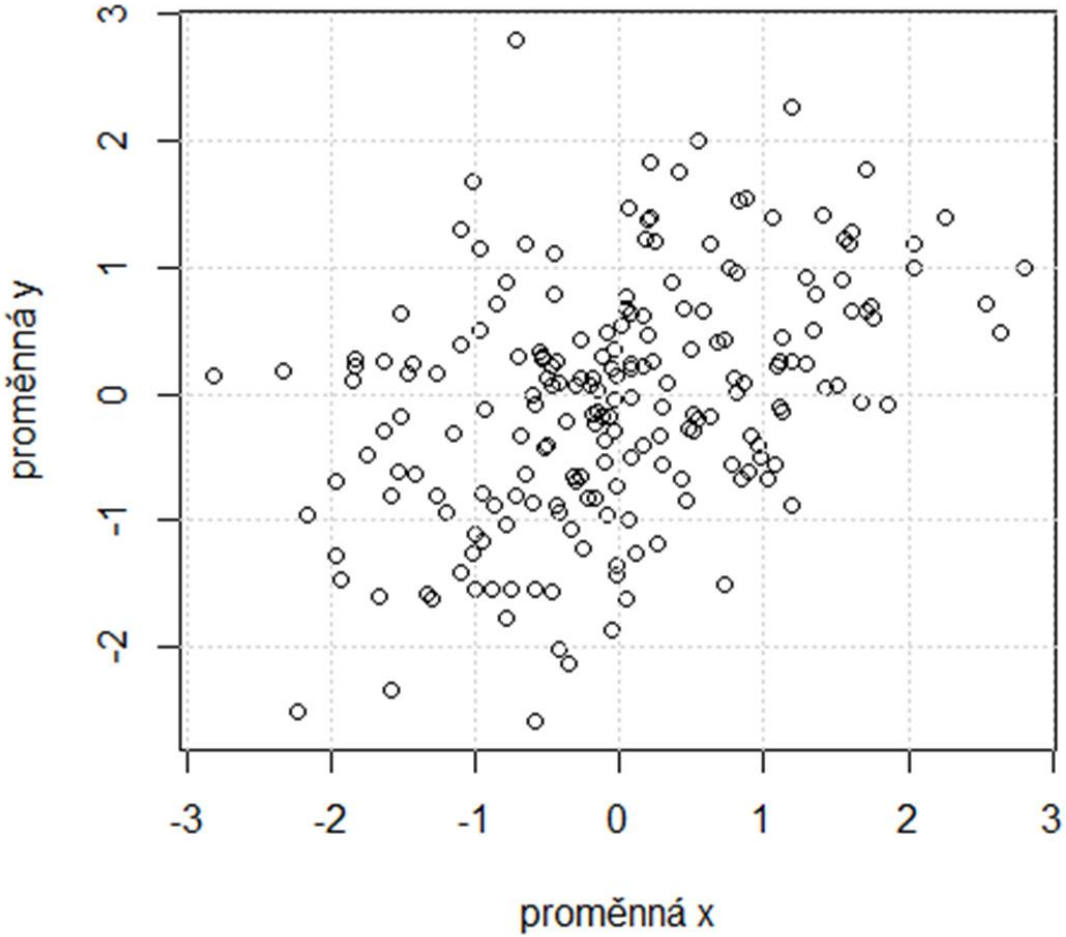
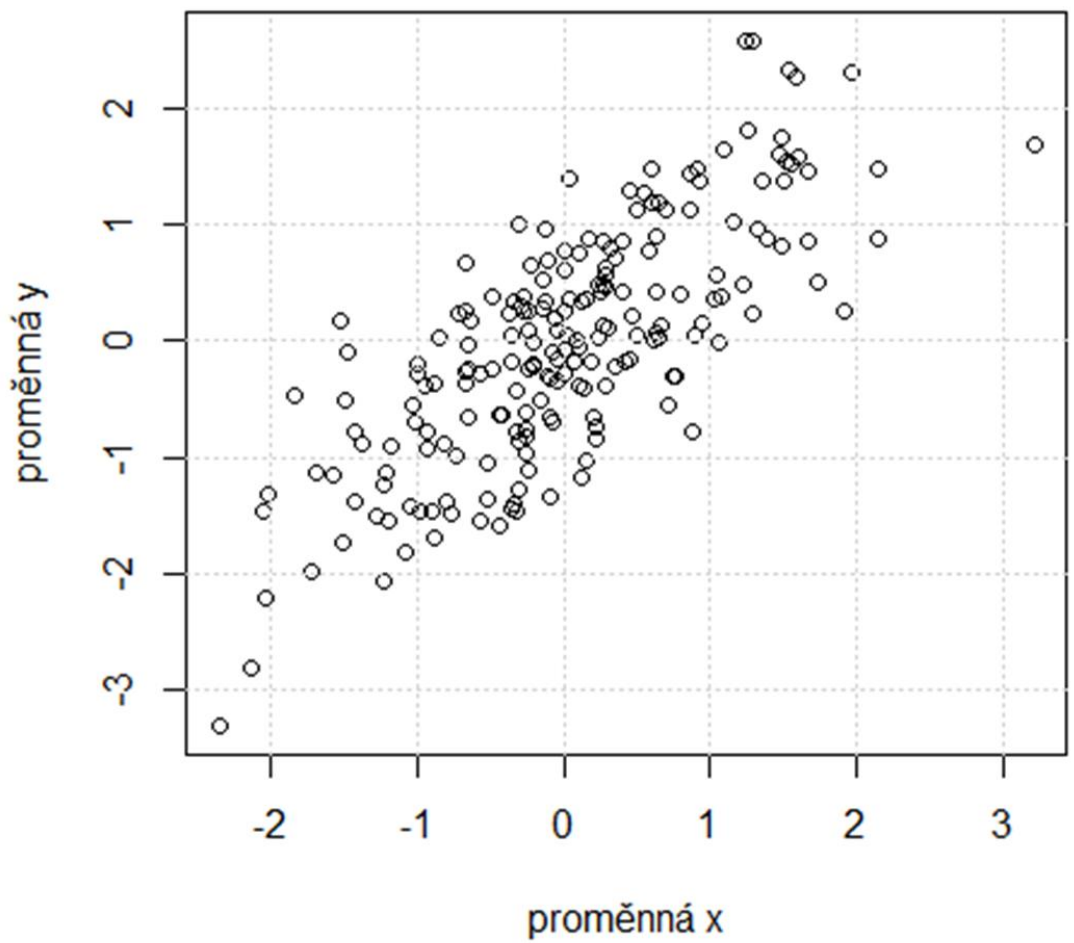
Data analysis with R & RStudio

Week 6 – correlation and regression

Tomáš Diviák
tomas.diviak@manchester.ac.uk


proměnná y
3
2
1
0
-1
-2
-3
-2
-1
0
1
2
3
proměnná x


proměnná y
3
2
1
0
-1
-2
-3
-2
-1
0
1
2
3
proměnná x


proměnná y
2
1
0
-1
-2
-3
-2
-1
0
1
2
3
proměnná x

Correlation

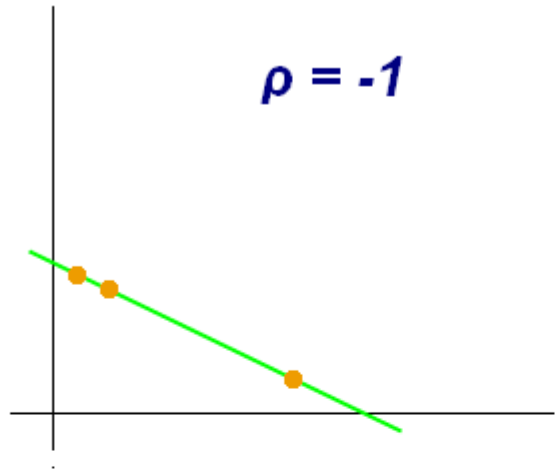
- Association between two numeric variables

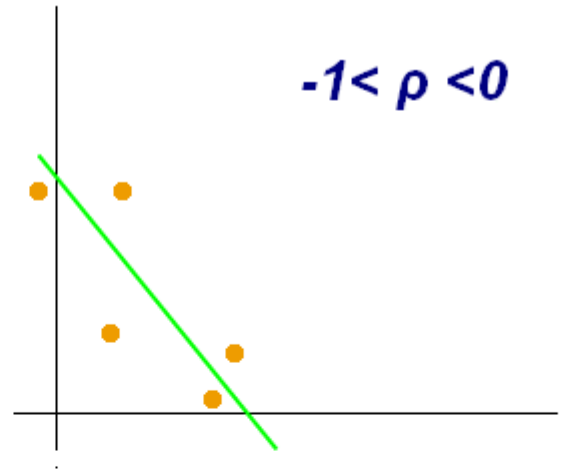
$$r = \frac{\sum (x - m_x)(y - m_y)}{\sqrt{\sum (x - m_x)^2 \sum (y - m_y)^2}}$$

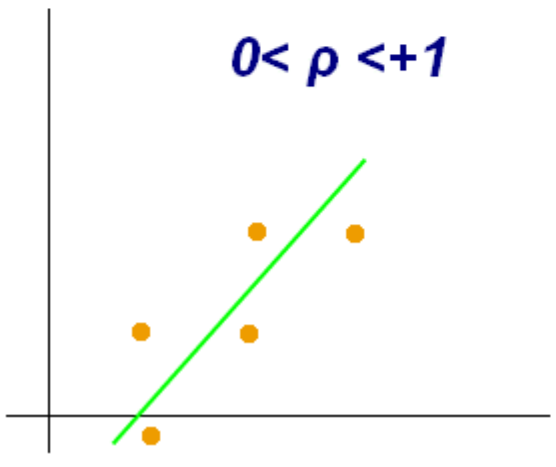
- **Correlation coefficients**: range **<-1; 1>**
  - **Pearson's product-moment correlation c.** – linear relationships
  - **Spearman's ρ** – for ranks
  - **Kendall's τ** – for ranks
  - **Interpretation**:
    - 0.0 – 0.2 / 0.3 – weak
    - Around 0.5 – moderate
    - > 0.8 – strong

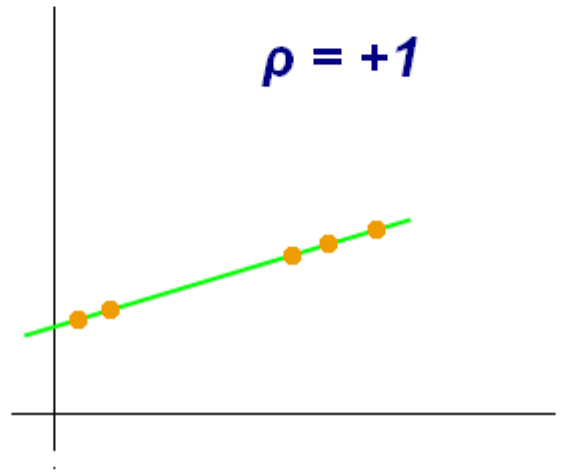
    →**only indicative**, better to consult previous research in a given field!

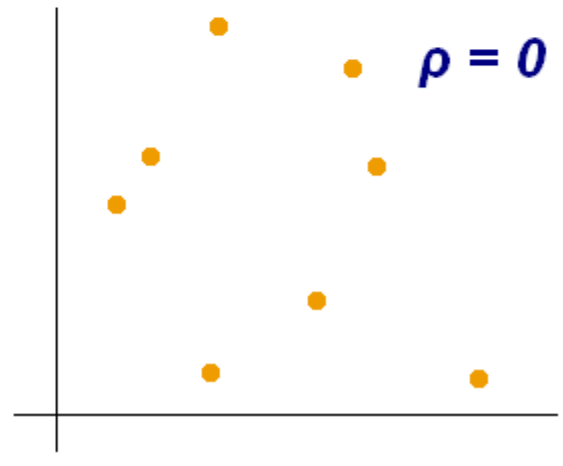
$$r_s = 1 - \frac{6\sum d^2}{n\left(n^2 - 1\right)}$$


ρ = -1


-1< ρ <0

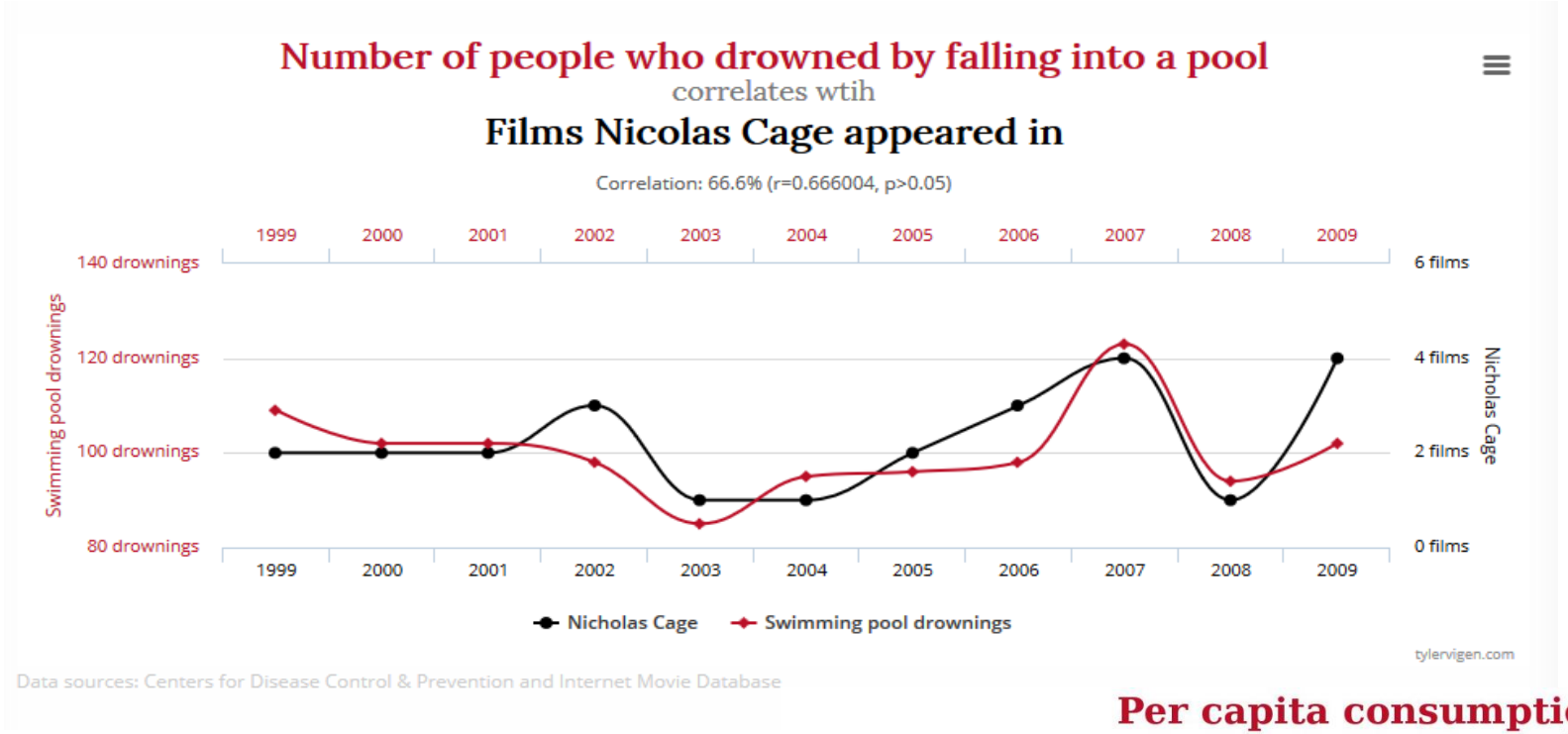

0< ρ <+1

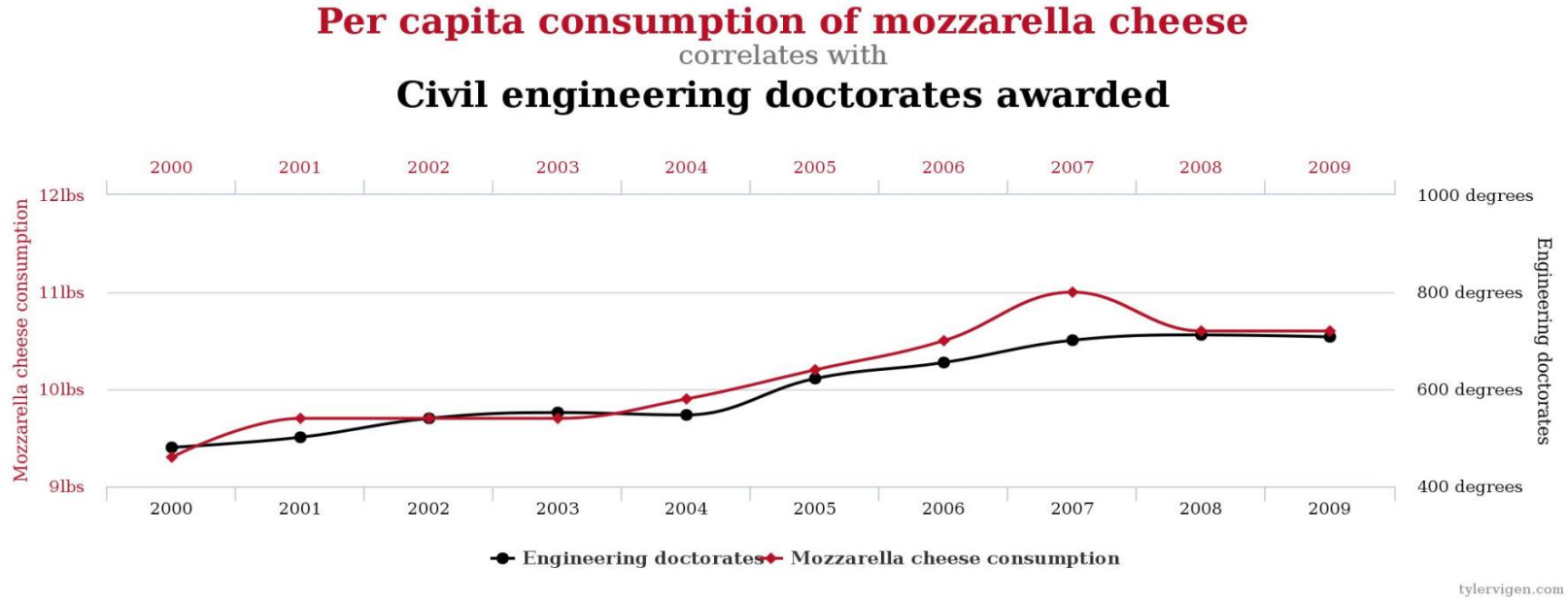

ρ = +1

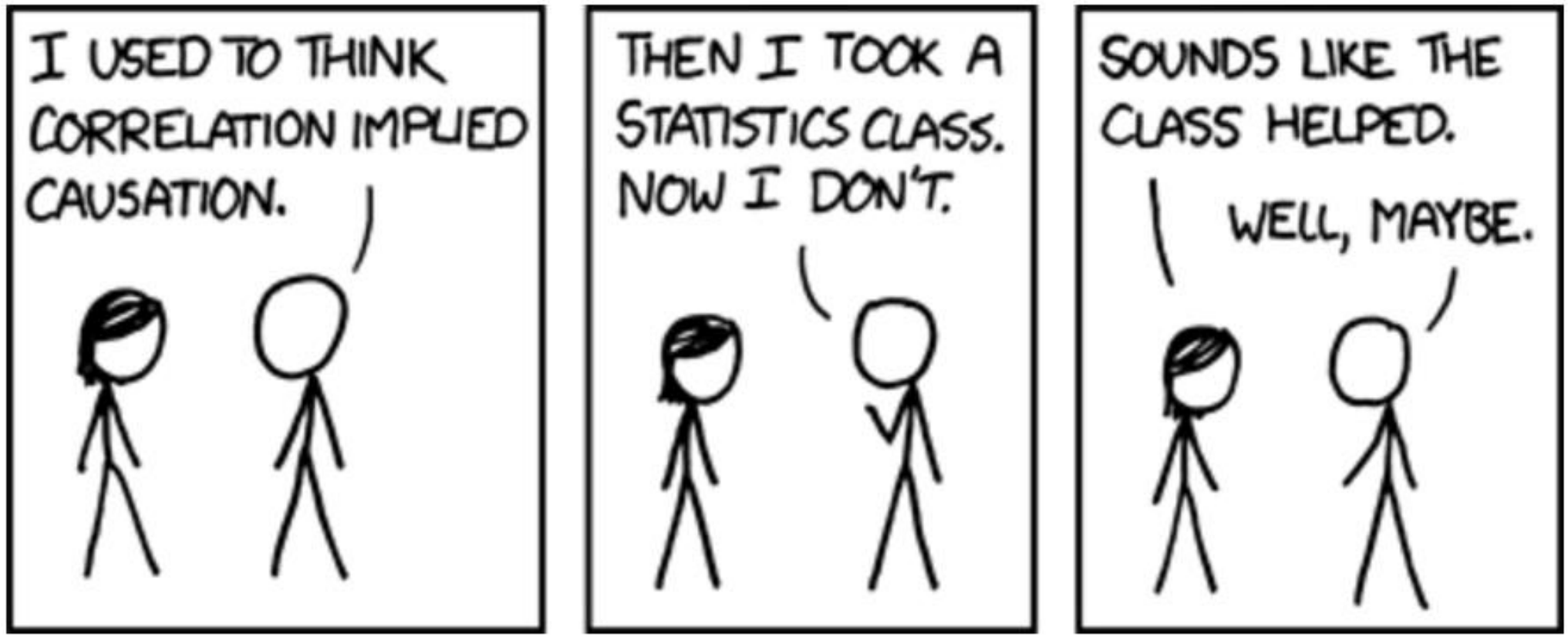

ρ = 0

Correlation and causation

- **CORRELATION DOES NOT IMPLY CAUSATION!!!**
- **Correlation is necessary, but not sufficient for causal claims**
  - **Temporal ordering** – A needs to always precede B in order to establish causality
  - **No effect of confounding variables** – the relationship between A and B should be adjusted for any potential confounding variable
- **Causal claims** thus require a good theory (causal mechanism), good methods (ideally experiments), and a lot of research to be backed up
- Example: fertility and stork population

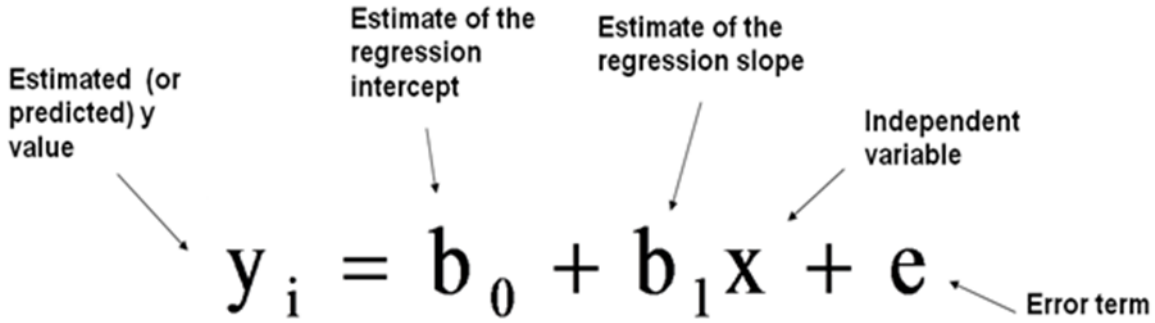

Number of people who drowned by falling into a pool
correlates wtih
Films Nicolas Cage appeared in
Correlation: 66.6% (r=0.666004, p>0.05)
1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009
140 drownings
120 drownings
100 drownings
80 drownings
Swimming pool drownings
6 films
4 films
2 films
0 films
Nicholas Cage
Nicholas Cage
Swimming pool drownings
tylervigen.com
Data sources: Centers for Disease Control & Prevention and Internet Movie Database

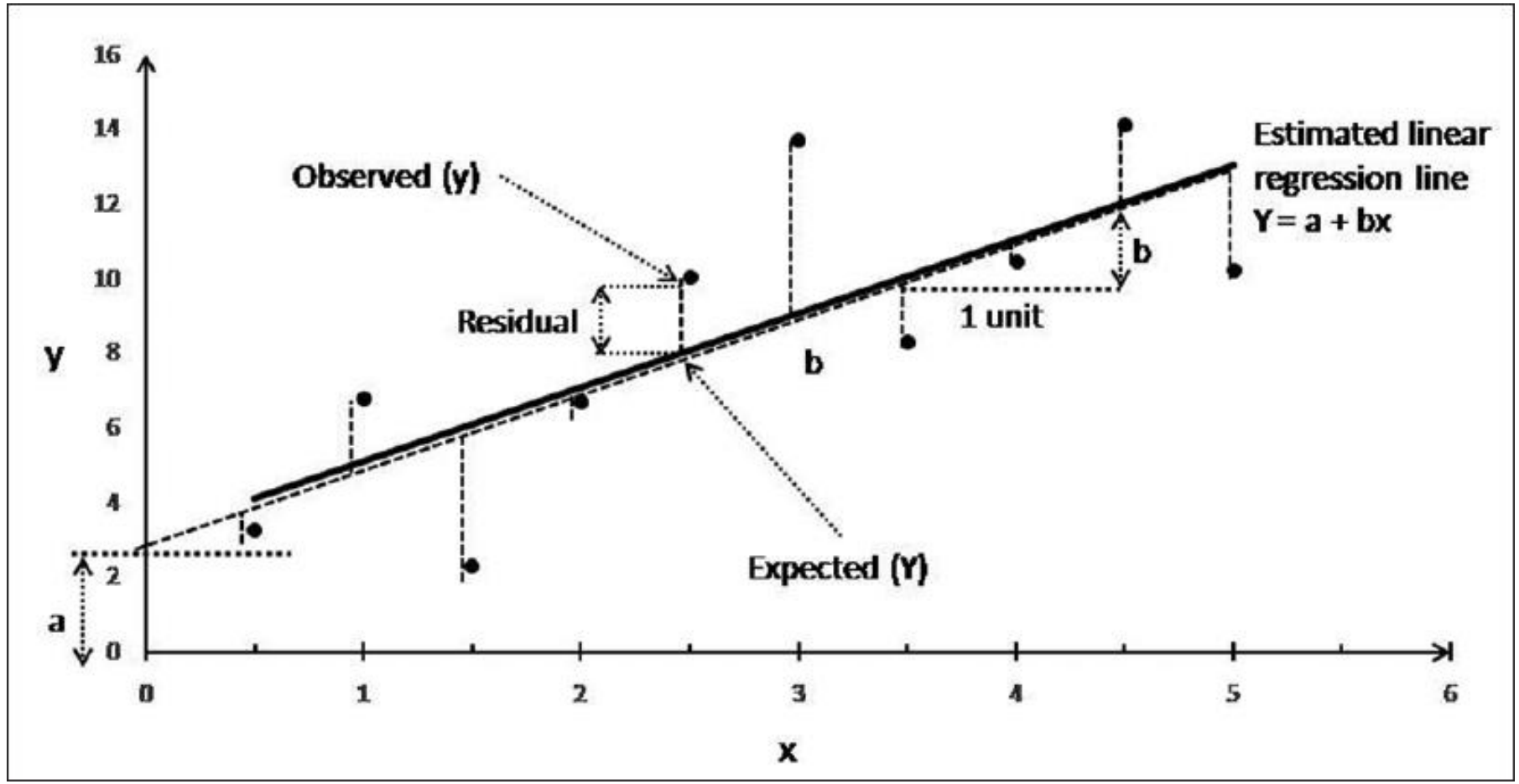

Per capita consumption of mozzarella cheese
correlates with
Civil engineering doctorates awarded
2000 2001 2002 2003 2004 2005 2006 2007 2008 2009
12lbs
11lbs
10lbs
9lbs
Mozzarella cheese consumption
1000 degrees
800 degrees
600 degrees
400 degrees
Engineering doctorates
Engineering doctorates
Mozzarella cheese consumption
tylervigen.com


I USED TO THINK CORRELATION IMPLIED CAUSATION.
THEN I TOOK A STATISTICS CLASS. NOW I DON'T.
SOUNDS LIKE THE CLASS HELPED.
WELL, MAYBE.

Linear regression

- **Predicting values of a numeric variable from values of another**
- **Outcome = predictor(s) + error** (or dependent variable = independent variable(s) + error)
  * error = some residual random fluctuation → we simplify from the world, which inherently comes with some error
- **Linear regression** – continuous outcome (y) = average (b_0) + (b_1x) continuous predictor (e.g., income = baseline + years of education)




16
14
12
10
8
6
4
2
0
y
a
0
1
2
3
4
5
6
x
Observed (y)
Residual
b
1 unit
b
Estimated linear
regression line
Y = a + bx
Expected (Y)

Linear regression

- Regression line aims to capture the variability in the data in the best way possible → **the method of ordinary least squares (OLS)**. The OLS finds a line that has the minimal sum of squared distances from the data points of all the possible lines.
- Regression line defined by $\mathbf{Y_i = B_0 + B_1 x_i + e_i}$
  - **Intercept $\mathbf{B_0}$ -** the point in which the line intersects with the Y axis → **the predicted value of Y if X = 0**
  - **Slope $\mathbf{B_1}$ -** may be positive or negative, **a value by which Y increases/decreases when we increase X by 1 unit**
- Squared distance of a data point from its prediction = **residual** (an error of prediction essentially)
- **Coefficient of determination ($\mathbf{R^2}$)** = proportion of the variability explained by the model the overall variability in the dependent variable (the more, the better the model, range <0; 1>)

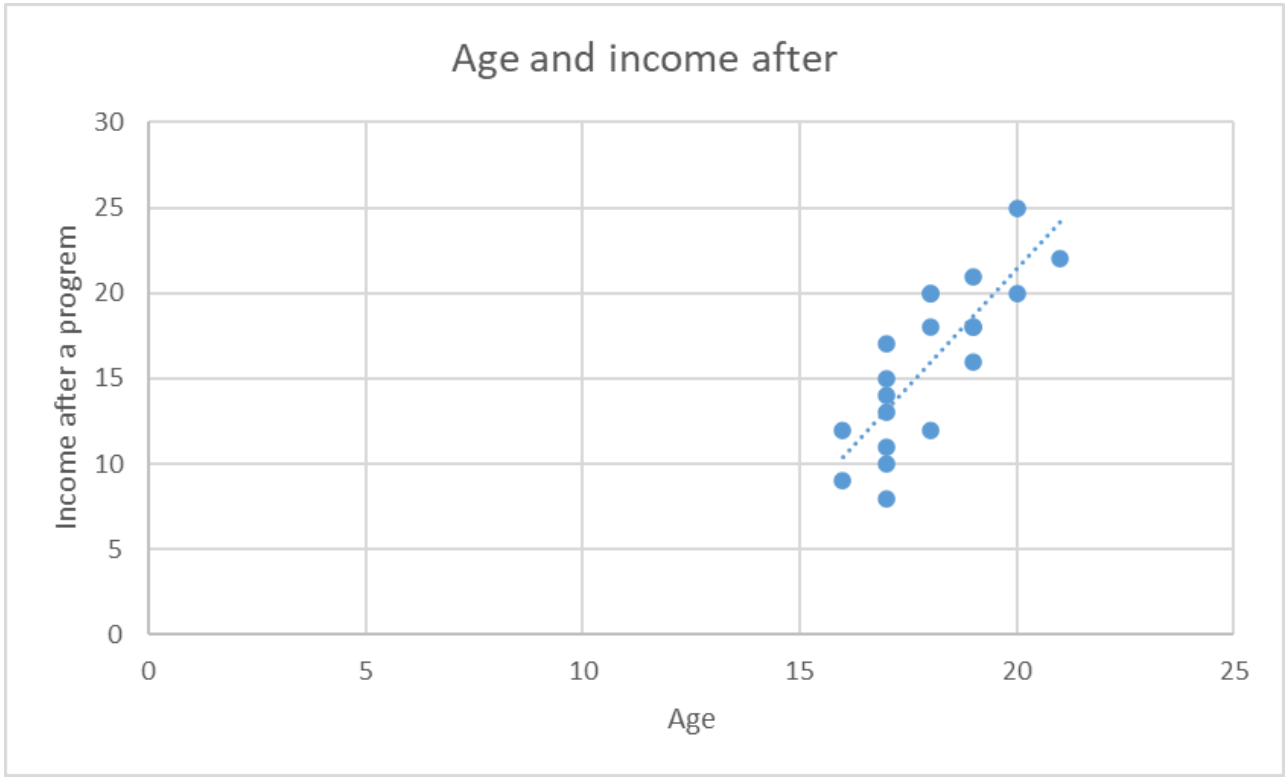
age	income after
16	12
17	10
17	11
19	18
18	12
17	15
17	13
21	22
18	18
16	9
17	8
18	20
17	14
19	16
20	25
18	20
19	18
19	21
20	20
17	17

Intercept B_0 **Slope B_1**

-33.55 2.75

Correlation

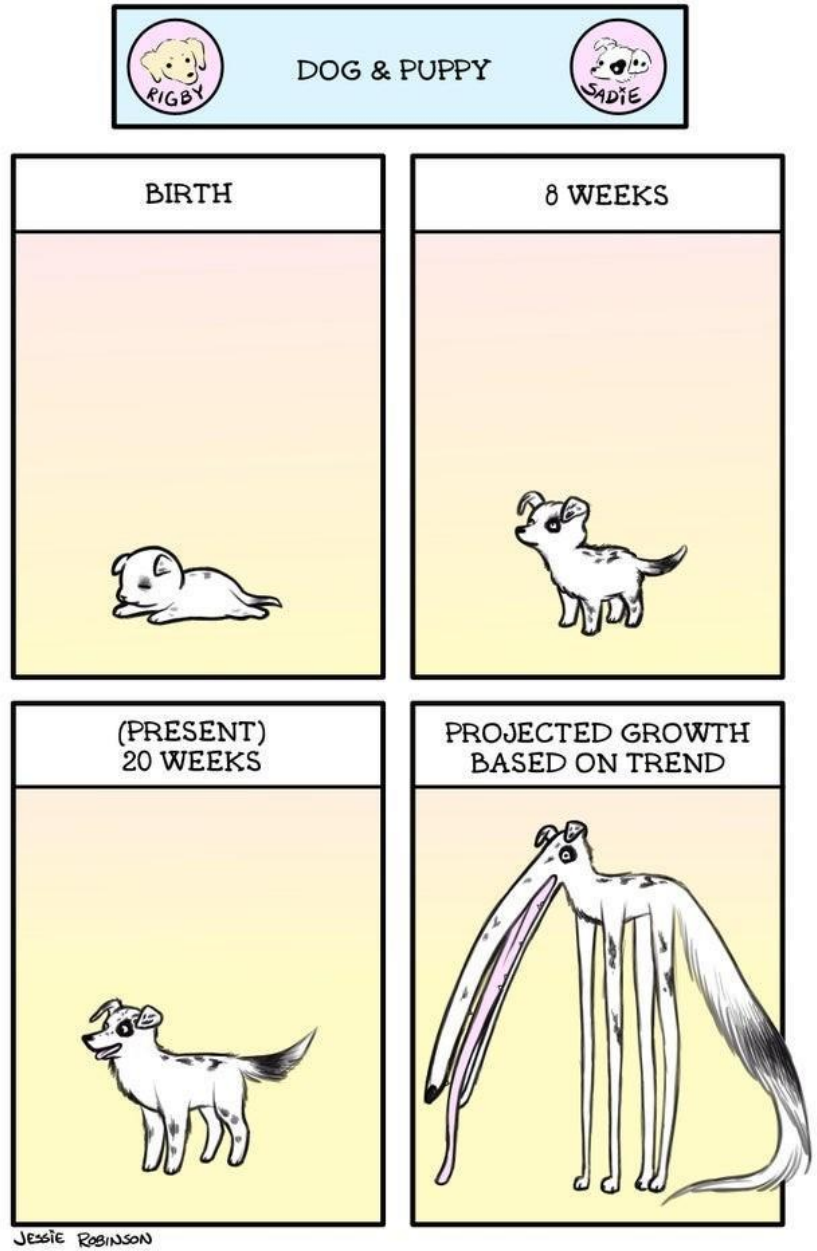
0.8



What would be the income of a 29-years old person? →

Income after = B_0 + B_1 * age →

<u>46.2</u> (income after) = -33.55 + 2.75 * 29 (age)


RIGBY
DOG & PUPPY
SADIE
BIRTH
8 WEEKS
(PRESENT)
20 WEEKS
PROJECTED GROWTH
BASED ON TREND
JESSIE ROBINSON

Thanks for your attention ☺

Questions?

